N5

NATIONAL

QUALIFICATIONS

Mark

N5/L

Mathematics

Combined Paper

45 minutes

Prelim Practice L

Fill in these boxes and read what is printed below

Forename(s)

Surname

Teacher

Total Marks - 30

Attempt ALL questions.

You must NOT use a calculator.

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this book to the Invigilator; if you do not, you may lose all the marks for this paper.

FORMULAE LIST

The roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$ or $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area of a triangle: $A = \frac{1}{2} ab \sin C$

Volume of a sphere: $V = \dfrac{4}{3}\pi r^3$

Volume of a cone: $V = \dfrac{1}{3}\pi r^2 h$

Volume of a pyramid: $V = \dfrac{1}{3} Ah$

Standard deviation: $s = \sqrt{\dfrac{\sum (x - \bar{x})^2}{n - 1}}$

or $s = \sqrt{\dfrac{\sum x^2 - \dfrac{(\sum x)^2}{n}}{n - 1}}$, where n is the sample size.

MARKS

Section A – NON-CALCULATOR
Total marks — 14
Attempt ALL questions

1. Determine the nature of the roots of the function $f(x) = 4x^2 - 3x + 2$. **2**

2. Given $\boldsymbol{e} = \begin{pmatrix} 3 \\ -7 \\ -2 \end{pmatrix}$ and $\boldsymbol{f} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$, find the components of the resultant vector $2\boldsymbol{f} + \boldsymbol{e}$. **2**

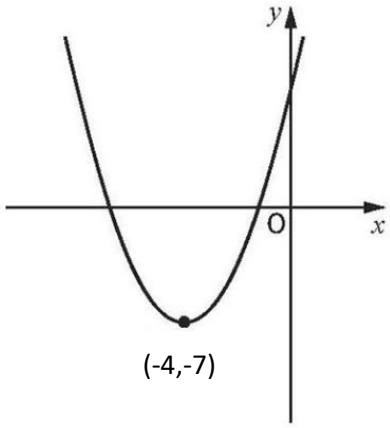
3. Express as a single fraction, where $x \neq 0, x \neq 3$:

$$\frac{5}{x-3} - \frac{2}{x}$$

2

4. Express $x^2 + 8x - 1$ in the form $(x + p)^2 + q$. **2**

5. Part of the graph of $y = (x + a)^2 + b$ is shown.



a) State the value of a. **1**

b) State the value of b. **1**

c) Determine the coordinates of the point where the parabola intercepts the y-axis. **2**

6. Simplify:

$$\frac{1 - \cos^2 x}{\cos^2 x}$$

2

[END OF SECTION A]

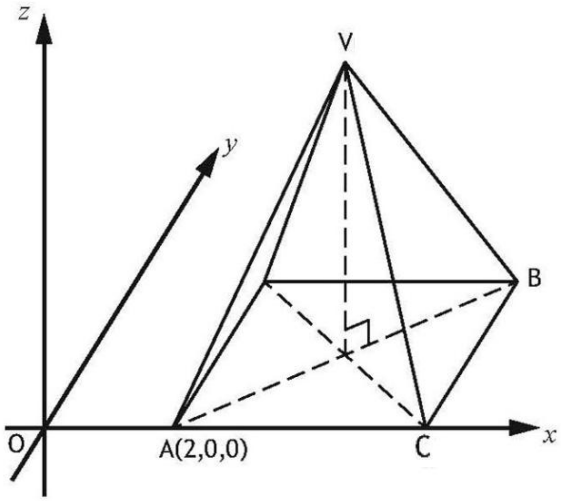
Section B – CALCULATOR ALLOWED
Total marks — 16
Attempt ALL questions

7. Solve the equation $2x^2 + 8x - 3 = 0$.
Give your answer correct to **one decimal place**. **3**

8. Given $\boldsymbol{w} = \begin{pmatrix} -2 \\ 2 \\ -4 \end{pmatrix}$, show that the magnitude, $|\boldsymbol{w}| = 2\sqrt{6}$. **3**

9. The diagram below shows a square-based pyramid.

- The height of the pyramid is 7 units.
- The distance from A to C is 4 units.
- Point V is vertically above the centre of the pyramid.



a) State the coordinates of point B **1**

b) State the coordinates of point V. **1**

10. The height of a part of a machine in a factoring whilst the machine is running can be described using the equation:

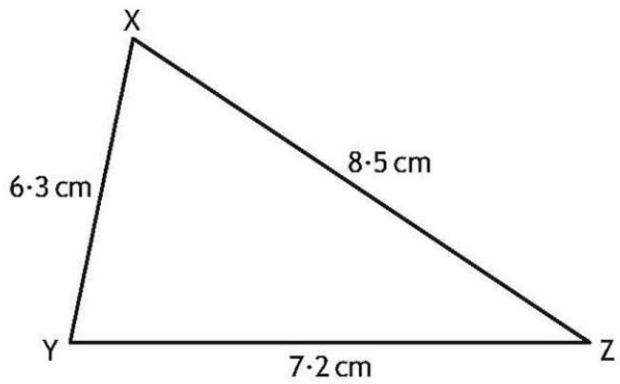
$$h = 6\sin t^\circ + 2$$

Where:

- h is height measured in metres.
- t is time since the machine was started, measured in seconds.

Calculate the first two times that the part is at a height of 7 metres, **4**

11. Triangle XYZ is shown.



Calculate the size of the largest angle in triangle XYZ. **4**

[END OF SECTION B]

ANSWERS

1. $-32 < 0$ so no real roots

2. $\begin{pmatrix} 1 \\ -3 \\ 6 \end{pmatrix}$

3. $\frac{3x+6}{x(x-3)}$

4. $(x+4)^2 - 17$

5. a) $a = 4$ b) $b = -7$ c) $(0,9)$

6. $\tan^2 x$

7. $x = 0.3, x = -4.3$

8. $\sqrt{(-2)^2 + 2^2 + (-4)^2} = \sqrt{24} = \sqrt{4}\sqrt{6} = 2\sqrt{6}$

9. a) $B(6,4,0)$ b) $V(4,2,7)$

10. After 56.4 seconds and 123.6 seconds

11. $77.7°$